

YUN-RONG (LAUREN) LUO

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RESEARCH INTERESTS

Distributed systems, formal verification, system performance, and programming languages

EDUCATION

University of Michigan, Ann Arbor (UMich)

Ph.D. Candidate in Computer Science and Engineering

Ann Arbor, MI

08/2023-Present

- Research Advisor: Prof. Manos Kapritsos
- Coursework: Distributed Systems, Formal Verification of Systems Software, Advanced Programming Languages, AI Foundations

National Taiwan University (NTU)

M.S. in Electronics Engineering

Taipei, Taiwan

09/2021-2023

- Research Advisor: Prof. Jie-Hong Roland Jiang
- Coursework: Advanced Algorithms, Graph Theory, Programming Languages, Physical Design for Nanometer Ics

B.S. in Electrical Engineering

09/2017-06/2021

- Coursework: Data Structures, Algorithms, Operating Systems, Embedded Systems Lab, Logic Synthesis and Verification, System-on-Chip Verification, Electronic Design Automation, Computing Theory, Complexity Theory, Mathematical Logic

PUBLICATIONS

Conference

1. **Yun-Rong Luo**, Aman Goel, Karem Sakallah, "QSM-Cutoff: Systematic Derivation of Quantified Cutoff Formulas for Distributed Protocols." 2025 International Conference on Computer Aided Verification (CAV), 2025 [[link](#)]
2. Che Cheng*, **Yun-Rong Luo***, Jie-Hong Roland Jiang, "Knowledge Compilation for Incremental and Checkable Stochastic Boolean Satisfiability." 2024 International Joint Conference on Artificial Intelligence (IJCAI), 2024 [[link](#)]
3. Yu-Neng Wang*, **Yun-Rong Luo***, Po-Chun Chien*, Ping-Lun Wang, Hao-Ren Wang, Wan-Hsuan Lin, Jie-Hong Roland Jiang, Chung-Yang Ric Huang, "Compatible Equivalence Checking of X-Valued Circuits." 2021 IEEE/ACM International Conference on Computer-Aided Design (ICCAD), 2021 [[link](#)]

Journal

1. **Yun-Rong Luo***, Che Cheng*, Jie-Hong Roland Jiang, "A Resolution Proof System for Dependency Stochastic Boolean Satisfiability." Journal of Automated Reasoning, 2023 [[link](#)]

RESEARCH EXPERIENCE

Graduate Student Research Assistant (UMich)

Performance Verification for Distributed Systems

Ann Arbor, MI

2025-Present

- Developing formal performance guarantees for distributed systems, including latency and throughput bounds.
- Applying stochastic network calculus to derive performance bounds for distributed systems.

Formal Verification for Distributed Protocols

2023-2025

- Developed [QSM-Cutoff](#): an automated inductive invariant inference tool for distributed protocols.
- Enhanced forward-reachability and inductive-invariant inference algorithms using symmetry and logic minimization.

Graduate Student Research Assistant (NTU)

Certified Stochastic Boolean Satisfiability Solving

Taipei, Taiwan

2023-2024

- Enabled proof generation for the stochastic Boolean satisfiability (SSAT) solver [SharpSSAT](#) via knowledge compilation.
- Extended a certified model-counting toolchain into [cert-SSAT](#) for certifying SharpSSAT results and helped identify a bug.

Proof System for Dependency Stochastic Boolean Satisfiability

2021-2023

- Developed the first sound and complete resolution proof system for dependency stochastic Boolean satisfiability (DSSAT).
- Formalized the DSSAT proof system and proved its soundness and completeness.

Undergraduate Student Researcher (NTU)

Equivalence Checking of X-Valued Circuits

Taipei, Taiwan

2020-2021

- Contributed to the first-place solution for *equivalence checking of X-valued circuits* at the 2020 ICCAD CAD Contest.
- Investigated SAT encodings for verification and optimized the circuit rewriting algorithm on 28 industrial cases.

WORK EXPERIENCE

Cadence Design Systems

R&D Intern, Formal Verification Team

Hsinchu, Taiwan

07/2020-09/2020

- Generated *engineering change order (ECO)* test cases for the Conformal ECO tool.

SELECTED PROJECTS

EECS 491: Introduction to Distributed Systems (UMich)

Winter 2026

- Implemented the following systems in Go:
 - a primary/backup key-value service, a Paxos-based key-value service, and a sharded key-value service with Paxos groups.

CSE 498/598: Formal Verification of Systems Software (UMich)

Fall 2025

- Verified a sharded key-value store in Dafny.

CSE 590: Advanced Programming Languages (UMich)

Winter 2025

- Implemented an automated test-input generator for C programs in OCaml as part of a two-person team.
- Implemented an automated random test-input generation technique based on the DART project.

TEACHING EXPERIENCE

EECS 376: Foundations of Computer Science (UMich)

Graduate Student Instructor

Spring 2025

EECS 270: Introduction to Logic Design (UMich)

Graduate Student Instructor

Fall 2024, Winter 2025

Logic Synthesis and Verification (NTU)

Teaching Assistant

09/2021-01/2022

HONORS AND AWARDS

2021 Research Creativity Award, Ministry of Science and Technology of Taiwan (awarded to 200/3000+ research works)

2020 1st Place, Problem A, 2020 CAD Contest at ICCAD (186 teams competed in 3 problems)

2020 College Student Research Scholarship, Ministry of Science and Technology of Taiwan

SKILLS

Programming Languages C/C++, Python, Go, OCaml, Verilog

Systems Linux, Shell scripting, Git, debuggers (gdb, pdb, ocamldebug), profilers (valgrind, perf, cProfile), $\text{\LaTeX}$

Verification Dafny, SAT/SMT solvers (Z3, CaDiCaL, Kissat, PySAT), Berkeley ABC Logic Synthesis and Verification System

Languages Mandarin (Native), English (Proficient, TOEFL: 114/120)